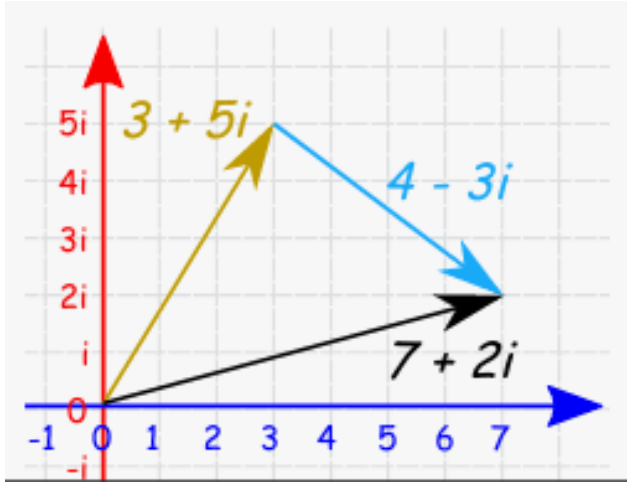
# COMPLEX ANALYSIS AND MATHEMATICAL TRANSFORMS
# IS 3302



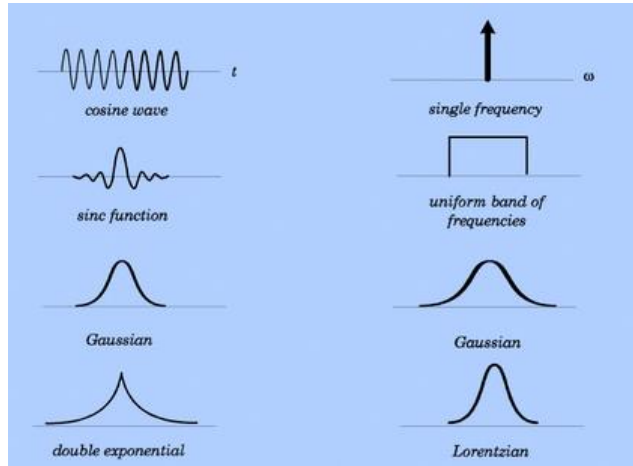

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# Content

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# Singular point

$z_0$ is a singular point of a complex function $f(z)$ if;

- $f(z)$ is not analytic in $z_0$.
- $f(z)$ is analytic in a point in every neighbourhood of $f(z)$.

If a function $f(z)$ is failed to be analytic at a point $z_0$, but it is analytic at some points in every neighbourhood of $f(z)$, then $z_0$ is called a singular point, or singularity of $f(z)$.

## Classifying Singularities

There are several types of singularities (points where $f(z)$ is not analytic) in the complex plane.

- **Isolated Singular points**
- **Non-Isolated Singular points**

Isolated Singular points

- **Removable Singular points**
- **Pole**
- **Essential Singular points**

# Classifying Singularities Cts…

## Isolated Singularity

$z_0$ is called Isolated Singular point if

- $f(z)$ is not analytic at $z_0$.
- $f(z)$ is analytic in the deleted neighbourhood of $z_0$.

(in this case the function can be expanded in Laurent's series)

Examples

$$f(z) = \frac{z}{(z-7)} \quad , \quad f(z) = \frac{z-3}{z(z-5)(z-2)} \quad , \quad f(z) = \frac{1}{e^{z-2}}$$

## Non-Isolated Singular Points

If $z_0$ is a singular points and there are other singular points in the neighborhood of $z_0$, then $z_0$ is non-isolated singular point.

## Removable Singular Points

Let $a$ is an isolated singular point of $f(z)$.

Where $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n}$ ; $R_1 < |z - z_0| < R_2$

If the principal part (i.e., the terms involving negative powers of $(z - z_0)$) contains no terms, $z_0$ is called removable singularity.

Example

$$f(z) = \frac{sinz}{z}$$

$$f(z) = \frac{sinz}{z} = \mathrm{z}\left(\mathrm{z} - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots\right) = (1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots)$$

# Singular Points ctd…

## Pole

$$\frac{b_1}{z-z_0}+\frac{b_2}{(z-z_0)^2}+\ldots+\frac{b_m}{(z-z_0)^m} \quad \text{and } b_m \neq 0$$

The principal part of the Laurent series has a finite number of terms (say $m$), then $z_0$ is called a pole of order $m$. (Note that if $b_1 = b_2 = \ldots = 0$ and $b_m \neq 0$, the pole is still of order m)

If $m = 1$, then $z_0$ is called a simple pole; if $m = 2$, then $z_0$ is called a double pole.

If we can find a positive integer $n$ such that $\lim_{z \to z_0} (z - z_0)^m f(z) = A \neq 0$, then $z = z_0$ is called a pole of order $m$.

Examples

1. $f(z) = \frac{z}{z-7}$; $z = 7$ *is a simple pole.*

2. $f(z) = z + \frac{1}{(z-3)^4} + \frac{3}{(z-1)^5}$ ; $z$ *is a principal part,* $z = 3$ *is a pole of order* $4, z = 1$ *is a pole of order* $5$.

# Singular Points…

## Essential Singular Points

If the principal part of the Laurent series has an infinite number of terms then $z = z_0$ is called an **isolated essential singularity** of f(z).

Example

$$f(z) = e^{\frac{1}{z-2}} = 1 + \frac{1}{z-2} + \frac{1}{(z-2)^2} + \cdots$$

***The principal part consists of infinite number of terms.***

$\boldsymbol{Hence, z = 2}$ is an essential singular point.

# Examples

1. $f(z) = \frac{1}{z}$
2. $f(z) = \frac{z}{(z+1)(z-3)}$
3. $f(z) = z^2 + z$
4. $f(z) = \frac{1}{\cos(z)}$
5. $f(z) = \frac{1}{\sin(\frac{\pi}{z})}$
6. $f(z) = |z|^2$
7. $f(z) = \frac{4z}{z^2-2z+2}$
8. $f(z) = \frac{z+2}{(z-2)}$
9. $f(z) = \frac{z}{(z-7)}$
10. $f(z) = \frac{z-3}{z(z-5)(z-2)}$
11. $f(z) = \frac{1}{e^{z-2}}$

# Examples

- Find the singular points of the following functions and categorize them.
  - $f(z) = \frac{1}{(z-2)^3}$
  - $f(z) = \frac{3z-2}{(z-1)^2(z+1)(z-4)}$
  - $f(z) = \frac{z}{(z^2+4)^2}$
  - $f(z) = \frac{z^8+z^4+2}{(z-1)^3(3z+2)^2}$

# SEQUENCE
# &
# SERIES

# Sequence

A sequence is any succession of numbers. A general sequence is denoted by

$$a_1, a_2, \dots, a_n, \dots$$

Example:

- Harmonic Sequence

$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots$ the $n^{th}$ term is denoted by $a_n = \frac{1}{n}$.

- Alternating harmonic sequence

$1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots,$ the $n^{th}$ term is denoted by $a_n = \frac{(-1)^{n+1}}{n}$.

# Sequence ctd…

## Convergent and Divergent of a Sequence

The sequence $a_1, a_2, \ldots, a_n, \ldots$ is said to be convergent if the limit of $a_n$ as n increases can be found.

i.e. $\lim_{n\to\infty} a_n$ exists.

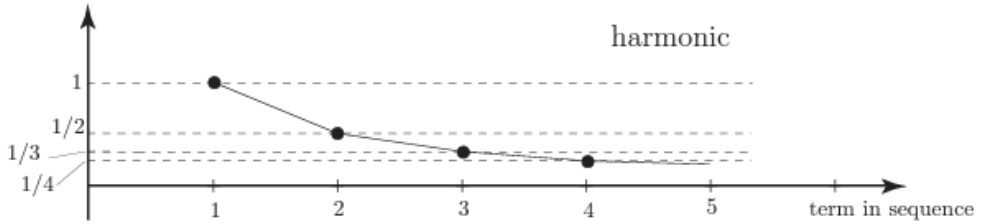
If the sequence is not convergent, it is said to be divergent.

Examples

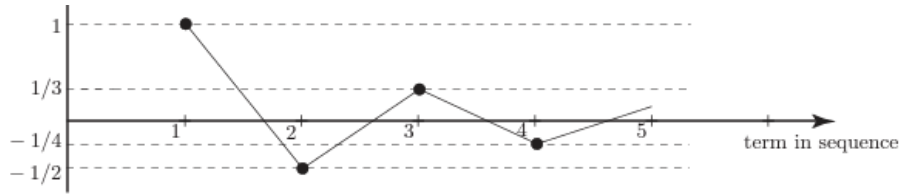
- Fibonacci sequence

  1, 1, 2, 3, 5, 8, ... Divergent

- Harmonic sequence

  1, 1 /2 , 1/ 3 , 1 /4 , ... 

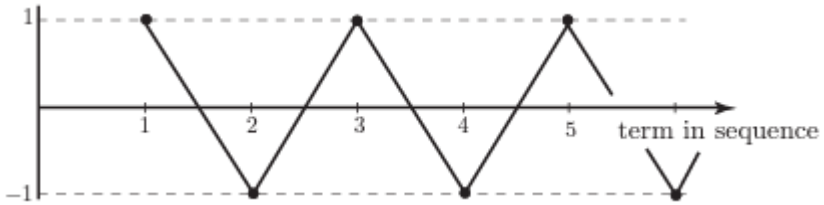
  Convergent

- 1, −1 /2 , 1 /3 , −1 /4 , ... 

  Convergent

# Sequence ctd…

- 1, −1, 1, −1, …




Not convergent/Oscillates

- $\frac{3}{1\times2}, \frac{4}{2\times3}, \frac{5}{3\times4}, \ldots$ $a_n = \frac{n+2}{n(n+1)}$ is convergent.

## Series

A series is the sum of the terms of a sequence. For the general sequence $a_1, a_2, \ldots, a_n, \ldots$ the series is $a_1 + a_2 + \cdots + a_n + \cdots$

- The sum of the first k terms is $\sum_{m=1}^{k} a_m$

## Arithmetic Series

$$[a] + [a + d] + [a + 2d] + \cdots + [a + (n-1)d]$$

If A is the sum of the first n terms; $A = \frac{1}{2}n[2a + (n-1)d]$

## Geometric Series

The geometric series $a + ar + ar^2 + \cdots + ar^{n-1}$ having n terms has sum $G$,

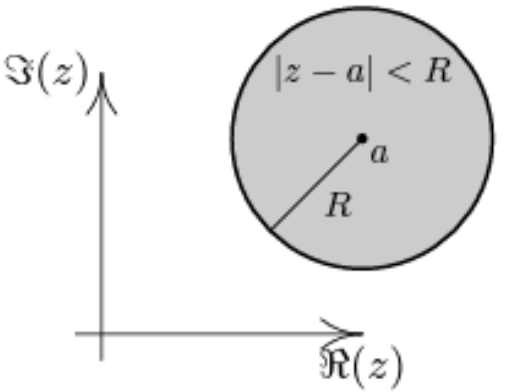
$G = \frac{a(1-r^n)}{1-r}$, if $r \neq 1$ and $G = na$, if $r = 1$.

# Power Series

A power series has the form

$\sum_{n=0}^{\infty} c_n(z-z_0)^n = c_0 + c_1(z-z_0) + c_2(z-z_0)^2 + \cdots$

where the $c_n$s are complex coefficients.




The power series converges if $|z-a| < R$

R is called the **radius of convergence** of the power series.

## Examples

- $Geometric\ Series = \sum_{k=0}^{\infty} z^k = 1 + z + z^2 + z^3 + \cdots = \frac{1}{1-z}$ for $|z| < 1$.

# The Ratio Test

convergence tests, applies only to series of **positive terms.**

Let $\sum_{p=1}^{\infty} a_p$ be a series of **positive** terms such that, as $p$ increases, the limit of $\frac{a_{p+1}}{a_p}$ equals a number $\lambda$. That is $\lim_{p\to\infty} \frac{a_{p+1}}{a_p} = \lambda$.

It can be shown that:

- if $\lambda > 1$, then $\sum_{p=1}^{\infty} a_p$ diverges
- if $\lambda < 1$, then $\sum_{p=1}^{\infty} a_p$ converges
- if $\lambda = 1$, then $\sum_{p=1}^{\infty} a_p$ may converge or diverge.

That is, the test is inconclusive in this case.

How to find the Radius of Convergence (R)

$$R = \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right|$$

Determining the distance from the center $z_0$ where the series converges, i.e.,$|z - z_0| < R$

# The Ratio Test

Use the ratio test to examine the convergence of the series

(a) $1+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\cdots$ (b) $1+x+x^2+x^3+\cdots$

$$1+\frac{1}{2!}+\frac{1}{3!}+\cdots=\sum_{p=1}^{\infty}\frac{1}{p!} \qquad a_p=\frac{1}{p!} \qquad \therefore \qquad a_{p+1}=\frac{1}{(p+1)!}$$

$$\lim_{p\to\infty}\frac{a_{p+1}}{a_p}=\lim_{p\to\infty}\frac{1}{(p+1)}=0$$

# Series…

## Taylor Series

Suppose a function $f$ is analytic throughout an open disk $|z - z_0| < R_0$ centered at $z_0$ with radius $R_0$. Then at each point $z$ in this disk $f(z)$ has the series representation

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \qquad \text{where } a_n = \frac{f^{(n)}(z_0)}{n!} \text{ for } (n = 0, 1, 2, \ldots)$$

In other words the function $f(z)$ can be represented exactly by the infinite series in the disk $|z - z_0| < R$

- A Taylor series is a power series that represents a **specific function** $f(x)$ ,centered at $z = z_0$.
- When $z_0 = 0$ the series is known as a **Maclaurin series**.

# Series…

- Here are some examples of well known Maclaurin series

$$
\begin{aligned}
e^z &= 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \ldots &&= \sum_{k=0}^{\infty} \frac{z^k}{k!} && |z| < \infty \\
\sin(z) &= z - \frac{z^3}{3!} + \frac{z^5}{5!} - \ldots &&= \sum_{k=0}^{\infty} \frac{(-1)^{k+1} z^{2k+1}}{(2k+1)!} && |z| < \infty \\
\cos(z) &= 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \ldots &&= \sum_{k=0}^{\infty} \frac{(-1)^{k+1} z^{2k}}{(2k)!} && |z| < \infty \\
\sinh(z) &= z + \frac{z^3}{3!} + \frac{z^5}{5!} + \ldots &&= \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)!} && |z| < \infty \\
\cosh(z) &= 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \ldots &&= \sum_{k=0}^{\infty} \frac{z^{2k}}{(2k)!} && |z| < \infty \\
\frac{1}{1-z} &= 1 + z + z^2 + z^3 + \ldots &&= \sum_{k=0}^{\infty} z^k && |z| < 1 \\
\ln(1+z) &= z - \frac{z^2}{2} + \frac{z^3}{3} + \ldots &&= \sum_{k=1}^{\infty} \frac{z^k}{k} && |z| < 1 \\
(1+z)^p &= 1 + pz + \frac{p(p-1)z^2}{2!} + \ldots \frac{p(p-1)\ldots(p-n+1)}{n!} \ldots && && |z| < 1 \\
\tan(z) &= z + \frac{z^3}{3} + \frac{2z^5}{15} + \ldots && && |z| < \tfrac{\pi}{2}
\end{aligned}
$$

# Examples

- If the function $g(z) = e^z$ is entire, then find the Taylor series expansion of $g(z)$ at $z = 0$.
- If the function $g(z) = \frac{1}{1+z}$ then find the Taylor series expansion of $g(z)$ at $z = 0$.
- If the function $g(z) = \frac{1}{1-z}; z \neq 1, |z| < 1$, then find the Taylor series expansion of $g(z)$ at $z = 0$.
- Find the Taylor series expansions of the following functions around zero.
  - $f(z) = cosz$
  - $f(z) = sinz$

# LAURENT SERIES

## Theorem

If $f(z)$ is analytic on $R_1 < |z - z_0| < R_2$, then

$$f(z) = \underbrace{\sum_{n=0}^{\infty} a_n \cdot (z - z_0)^n}_{\text{analytic part}} + \underbrace{\sum_{n=1}^{\infty} a_{-n} \cdot (z - z_0)^{-n}}_{\text{principal part}}, \text{ for all } z \text{ on } R_1 < |z - z_0| < R_2$$

Where;

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(s)}{(s - z_0)^{n+1}} ds \quad (n = 0, \pm 1, \pm 2, \ldots)$$

$$f(z) = \ldots + \frac{a_{-3}}{(z - z_0)^3} + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{(z - z_0)^1} + a_0 + a_1 \cdot (z - z_0)^1 + a_2 \cdot (z - z_0)^2 + a_3 \cdot (z - z_0)^3 + \ldots$$

# Series…

- Laurent Series

In fact, the best way to identify an essential singularity $z_0$ of a function $f(z)$ (and an alternative way to compute residues) is to look at the series representation of the function $f(z)$ about the point $z_0$
That is,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n}, \qquad R_1 < |z - z_0| < R_2$$

This formula for a Laurent series is sometimes written as

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n \qquad \text{where } c_n = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz, \quad n = \pm 1, \pm 2, \ldots$$

# Series…

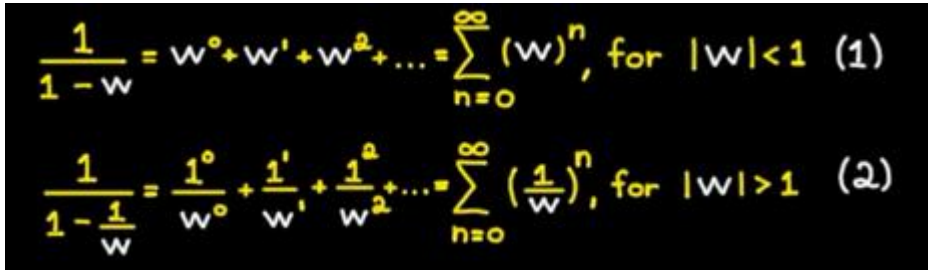
- Laurent Series…

- This first part of this series is clearly a **Taylor series** if bn = 0 for all n.

- This first part of the series representation is known as the **analytic part** of the function.

- The second part (with the negative exponents) is called the **principal part** of the function. However if $a_n$ and $b_n$ are not all identically zero this type of series is called a **Laurent series** and converges to the function f(z) in the annular region $R1 < |z - z_0| < R2$.

- Determine all possible Laurent series for each function.
- 1. $f(z) = \frac{1}{z-1}$ around $z = 0$




**Summary of Laurent series around $z = 0$:**

1. For $|z| < 1$:

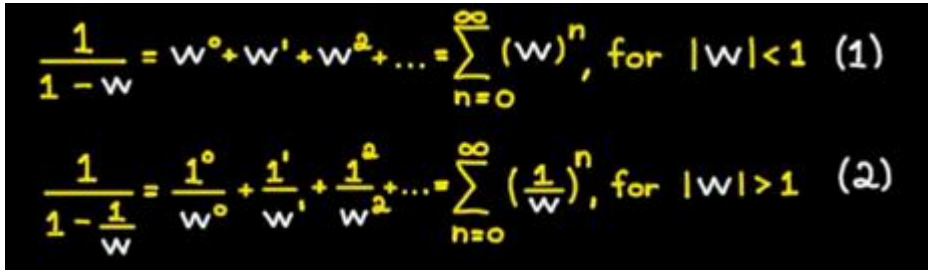
$$f(z) = -1 - z - z^2 - z^3 - \cdots = -\sum_{n=0}^{\infty} z^n$$

2. For $|z| > 1$:

$$f(z) = z^{-1} + z^{-2} + z^{-3} + \cdots = \sum_{n=1}^{\infty} z^{-n}$$

- Determine all possible Laurent series for each function.

2. $f(z) = \dfrac{1}{(z-1)(z-2)}$ around z = 0




**Summary of Laurent expansions around $z = 0$:**

- For $|z| < 1$:

$$f(z) = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right) z^n$$

- For $1 < |z| < 2$:

$$f(z) = -\sum_{n=0}^{\infty} z^{-(n+1)} - \frac{1}{2}\sum_{n=0}^{\infty} \frac{z^n}{2^n}$$

- For $|z| > 2$:

$$f(z) = \sum_{n=0}^{\infty} (2^n - 1) z^{-(n+1)}$$

- Determine all possible Laurent series for each function.

- 3. Expand $f(z) = \dfrac{1}{2-z}$ in terms of negative powers of $z$ to give a series which will be valid if $|z| > 2$.

$$\frac{1}{2-z} = -\left(\frac{1}{z} + \frac{2}{z^2} + \frac{4}{z^3} + \frac{8}{z^4} + \cdots\right), \quad |z| > 2$$

- Determine all possible Laurent series expansions of the function $f(z) = \frac{1}{z-2}$ ; $z \neq 2$ at $z_0 = 0$.

$$\frac{1}{z-2} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}, \quad |z| < 2,$$

$$\frac{1}{z-2} = \sum_{n=0}^{\infty} \frac{2^n}{z^{n+1}}, \quad |z| > 2.$$

- Determine all possible Laurent series expansions of the function $f(z) = \frac{1}{z-2}$ ; $z \neq 2$ at $z_0 = 1$.

$$\frac{1}{z-2} = -\sum_{n=0}^{\infty}(z-1)^n, \quad |z-1| < 1,$$

$$\frac{1}{z-2} = \sum_{n=0}^{\infty}\frac{1}{(z-1)^{n+1}}, \quad |z-1| > 1.$$

# Examples

- Expand $f(z) = \frac{1}{1-z}$ in terms of negative powers of $z$ which will be valid if $|z| > 1$.

$$f(z) = -\frac{1}{z}\sum_{n=0}^{\infty}\left(\frac{1}{z}\right)^n = -\sum_{n=0}^{\infty}\frac{1}{z^{n+1}}$$

$$\boxed{\frac{1}{1-z} = -\left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right), \qquad |z| > 1}$$

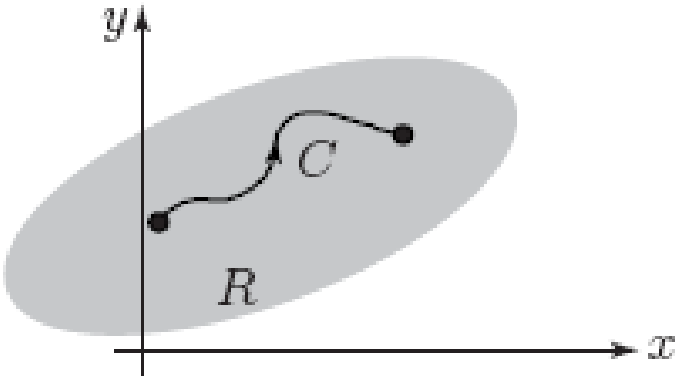
- Complex Integration
  - Cauchy's Integral Theorem
  - Cauchy's Integral Formula
  - Residue Theory

# Complex Integrals

## Basic Complex Integration

If f(z) is a single-valued, continuous function in some region R in the complex plane then we define the integral of f(z) along a path C in R as

$$\int_C f(z)dz = \int_C (u + iv)(dx + idy)$$




Here we have written f(z) and dz in real and imaginary parts:

$f(z) = u + iv$ and $dz = dx + idy$

Then we can separate the integral into real and imaginary parts as

$$\int_C f(z)dz = \int_C (udx - vdy) + i\int_C (vdx\, udy)$$

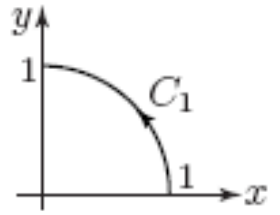
**Here we define complex integrals in terms of line integrals over paths in the complex plane.**

# Complex Integrals…

Example

- Obtain the complex integral
  - $\int_C zdz$ where $C$ is the straight line path from $z = 1 + i$ to $z = 3 + i$. (ans. 4+2i)
  - $\int_{C_1} zdz$ where $C_1$ is the straight line path from $z = 3 + i$ to $z = 3 + 3i$. (ans. -4+6i)
  - $\int_{C_2} zdz$ where $C_2$ is the straight line path from $z = 1 + i$ to $z = 3 + 3i$. (ans. 8i)

- Evaluate $\int_{C_1} z^2 dz$ where $C_1$ is that part of the unit circle going anticlockw
  point $z = 1$ to the point $z = i$. (ans. –(1+i)/3)



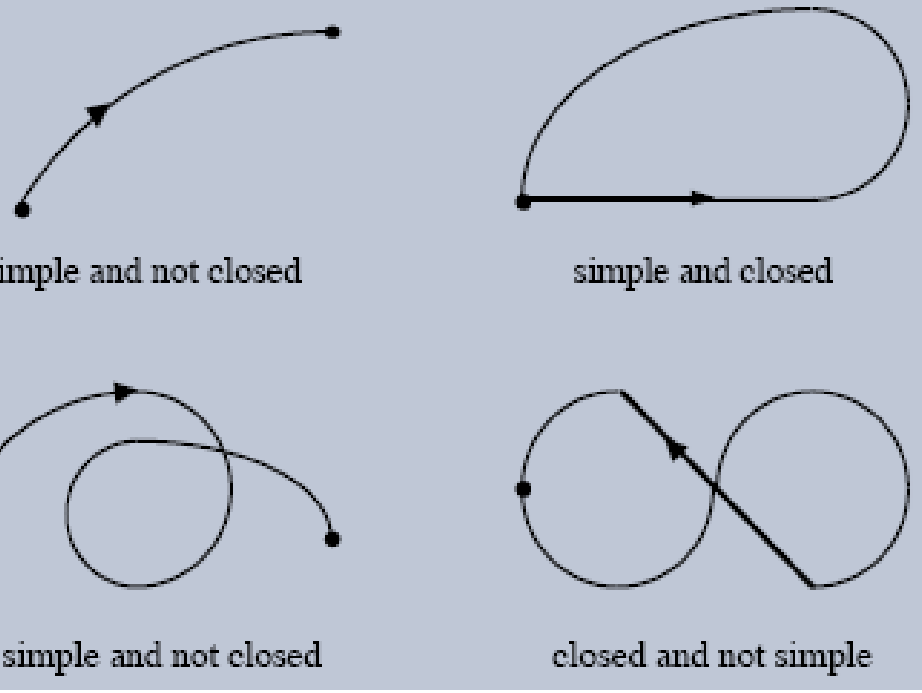

- Evaluate $\int_C \frac{1}{z} dz$ where $C$ is the unit circle.

# Complex Integrals…

In this Section we consider **contour integrals**.

DEFINITION. A complex valued function $\zeta : [A, B] \to \mathbb{C}$ is called a curve. The curve is said to be continuous if $\zeta$ is continuous in $[A, B]$, and differentiable if $\zeta$ is also differentiable in $[A, B]$. The set $\zeta([A, B])$ is called the trace of the curve. The point $\zeta(A)$ is called the initial point of the curve, and the point $\zeta(B)$ is called the terminal point of the curve.

DEFINITION. A curve $\zeta : [A, B] \to \mathbb{C}$ is said to be simple if $\zeta(t_1) \neq \zeta(t_2)$ whenever $t_1 \neq t_2$, with the possible exception that $\zeta(A) = \zeta(B)$. A curve $\zeta : [A, B] \to \mathbb{C}$ is said to be closed if $\zeta(A) = \zeta(B)$.



# Contour Integrals

DEFINITION. A curve $\zeta : [A, B] \to \mathbb{C}$ is said to be an arc if $\zeta$ is differentiable in $[A, B]$ and $\zeta'$ is continuous in $[A, B]$.

EXAMPLE 4.2.1. The unit circle is a simple closed arc, since we can described it by $\zeta : [0, 2\pi] \to \mathbb{C}$, given by $\zeta(t) = e^{it} = \cos t + i \sin t$. It is easy to check that $\zeta(t_1) \neq \zeta(t_2)$ whenever $t_1 \neq t_2$, the only exception being $\zeta(0) = \zeta(2\pi)$. Furthermore, $\zeta'(t) = -\sin t + i \cos t$ is continuous in $[0, 2\pi]$.

DEFINITION. Suppose that $C$ is an arc given by the function $\zeta : [A, B] \to \mathbb{C}$. A complex valued function $f$ is said to be continuous on the arc $C$ if the function $\psi(t) = f(\zeta(t))$ is continuous in $[A, B]$. In this case, the integral of $f$ on $C$ is defined to be

$$\int_C f(z)\,dz = \int_A^B f(\zeta(t))\zeta'(t)\,dt. \tag{1}$$

REMARKS. (1) Note that (1) can be obtained by the formal substitution $z = \zeta(t)$ and $dz = \zeta'(t)\,dt$.

# Contour Integrals…

(2) If we describe the arc $C$ in the opposite direction from $t = B$ to $t = A$, this opposite arc can be designated by $-C$. Since

$$\int_B^A f(\zeta(t))\zeta'(t)\,\mathrm{d}t = -\int_A^B f(\zeta(t))\zeta'(t)\,\mathrm{d}t,$$

we have

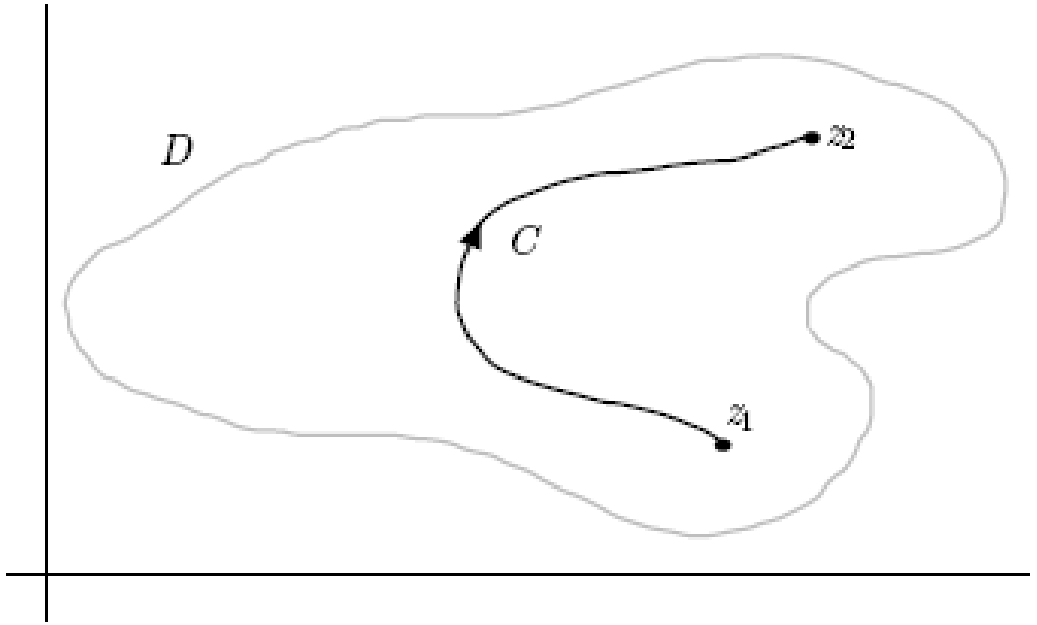
$$\int_{-C} f(z)\,\mathrm{d}z = -\int_C f(z)\,\mathrm{d}z.$$

(3) Integration of functions on arcs is a linear operation. More precisely, suppose that $f$ and $g$ are continuous on the arc $C$. Then for any $\alpha, \beta \in \mathbb{C}$, we have

$$\int_C (\alpha f(z) + \beta g(z))\,\mathrm{d}z = \alpha \int_C f(z)\,\mathrm{d}z + \beta \int_C g(z)\,\mathrm{d}z.$$

# Contour Integrals…

(4) Suppose that $F$ is analytic in a domain $D$ and has a continuous derivative $f = F'$ in $D$. Suppose further that $C$, defined by $\zeta : [A, B] \to \mathbb{C}$, is an arc lying in $D$, with initial point $z_1$ and terminal point $z_2$.




Define $\psi : [A, B] \to \mathbb{C}$ by $\psi(t) = F(\zeta(t))$. Then by the Fundamental theorem of integral calculus applied to the function $\psi'(t) = F'(\zeta(t))\zeta'(t) = f(\zeta(t))\zeta'(t)$, we have

$$\int_C f(z)\,\mathrm{d}z = \int_A^B f(\zeta(t))\zeta'(t)\,\mathrm{d}t = F(\zeta(B)) - F(\zeta(A)) = F(z_2) - F(z_1).$$

# Contour Integrals…

DEFINITION. Suppose that $A_1 < B_1 = A_2 < B_2 = \ldots = A_k < B_k$ are real numbers, and that for every $j = 1, \ldots, k$, $C_j$ is an arc given by the function $\zeta_j : [A_j, B_j] \to \mathbb{C}$. Suppose further that $\zeta_j(B_j) = \zeta_{j+1}(A_{j+1})$ for every $j = 1, \ldots, k-1$. Then

$$C = C_1 \cup C_2 \cup \ldots \cup C_k$$

is called a contour. The point $\zeta_1(A_1)$ is called the initial point of the contour $C$, and the point $\zeta_k(B_k)$ is called the terminal point of the contour $C$. A complex valued function $f$ is said to be continuous on the contour $C$ if it is continuous on the arc $C_j$ for every $j = 1, \ldots, k$. In this case, the integral of $f$ on $C$ is defined to be

$$\int_C f(z)\,\mathrm{d}z = \int_{C_1} f(z)\,\mathrm{d}z + \int_{C_2} f(z)\,\mathrm{d}z + \ldots + \int_{C_k} f(z)\,\mathrm{d}z.$$

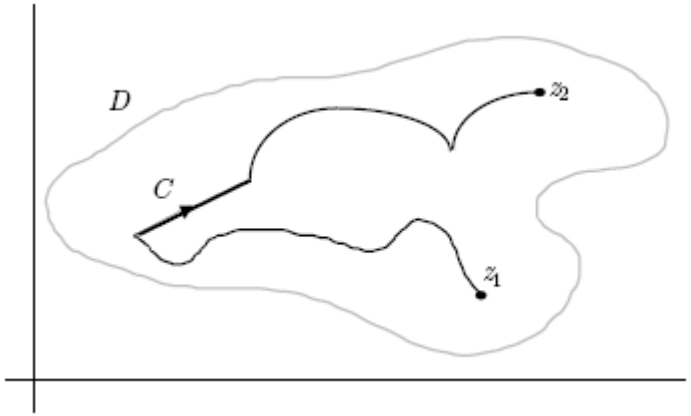
## Contour Integrals…

Theorem

*Suppose that F is analytic in a domain D and has a continuous derivative f = F' in D. Suppose further that C is a contour lying in D, with initial point z1 and terminal point z2.*

*Then,*

$$\int_C f(z)\,dz = F(z_2) - F(z_1).$$




Note that the right hand side of the above equation is independent of the contour *C*. It follows that under the hypotheses of the Theorem above,

we have

$$\int_C f(z)\,dz = 0$$

for any closed contour *C* in *D*.

## Cauchy's Integral Theorem

Suppose that a function f(z) is analytic everywhere within a simply-connected region then:

$$\oint_C f(z)dz = 0$$

for every simple closed path C lying in the region.

Examples

$\int_C z^2 dz = 0$ ; where C is the unit circle.

**Simply-connected Region**

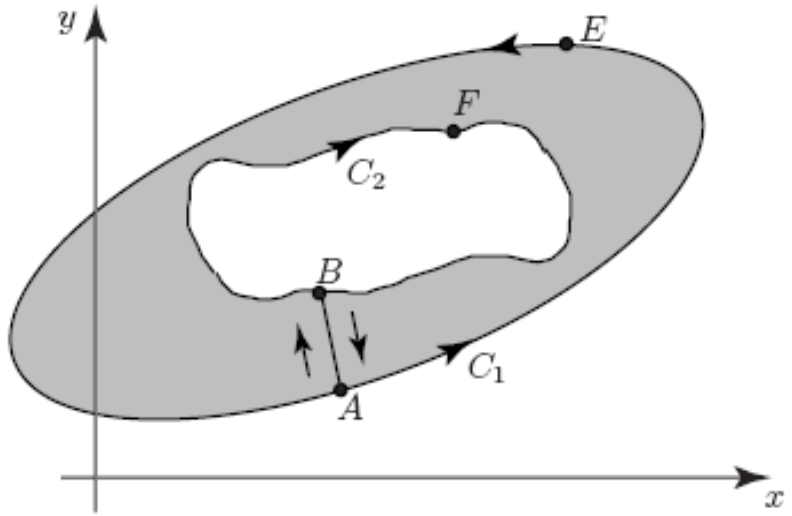
A region is said to be simply-connected if any closed curve in that region can be shrunk to a point without any part of it leaving a region.

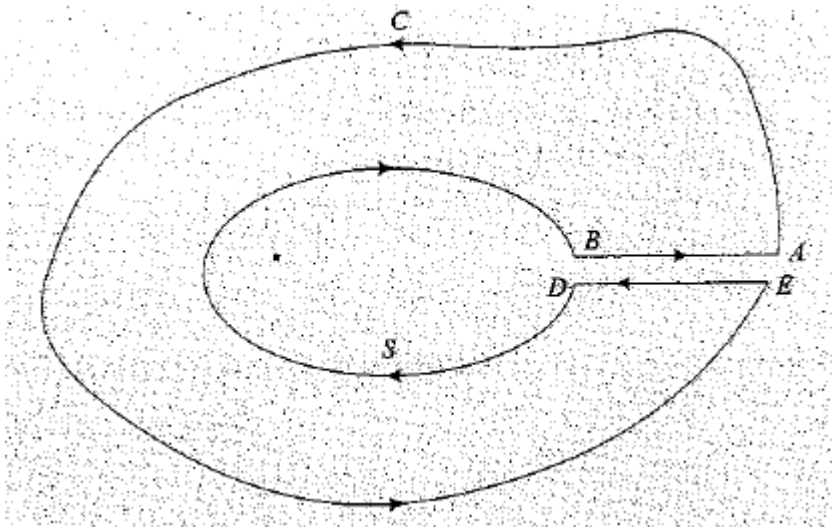


# CAUCHY THEOREM for Doubly connected Region

If $f(z)$ is analytic in a domain that contains two simple closed curves S and C, and the entire region lying between them, then, assuming they are oriented in the same direction,

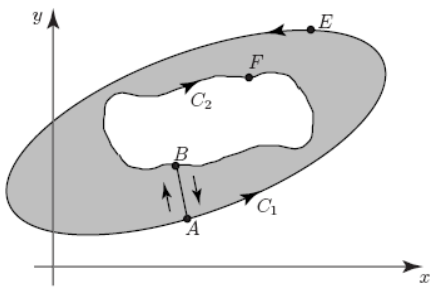
$$\oint_C f(z)dz = \oint_S f(z)dz$$



# Proof - CAUCHY THEOREM for Doubly connected Region

If $f(z)$ is analytic in a domain that contains two simple closed curves S and C, and the entire region lying between them, then, assuming they are oriented in the same direction,




$\oint_{AEABFBA} f(z)\,dz = 0$ since $f(z)$ is analytic within and on the curve $AEABFBA$.

Note that

$$\int_{AB} f(z)\,dz = -\int_{BA} f(z)\,dz,$$ being a simple change of direction.

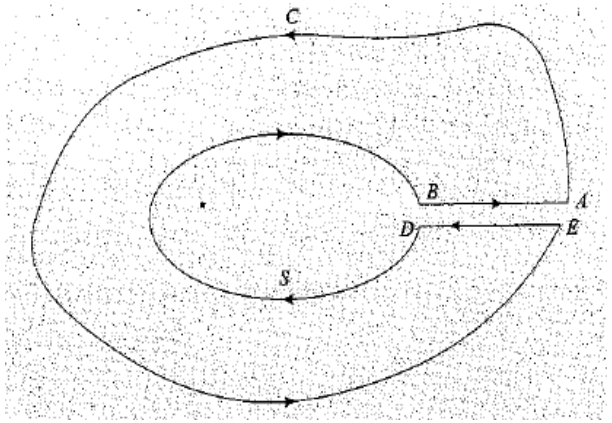
Also, we can divide the closed curve into smaller sections:

$$\oint_{AEABFBA} f(z)\,dz = \int_{AEA} f(z)\,dz + \int_{AB} f(z)\,dz + \int_{BFB} f(z)\,dz + \int_{BA} f(z)\,dz$$

$$= \int_{AEA} f(z)\,dz + \int_{BFB} f(z)\,dz = 0.$$

i.e.

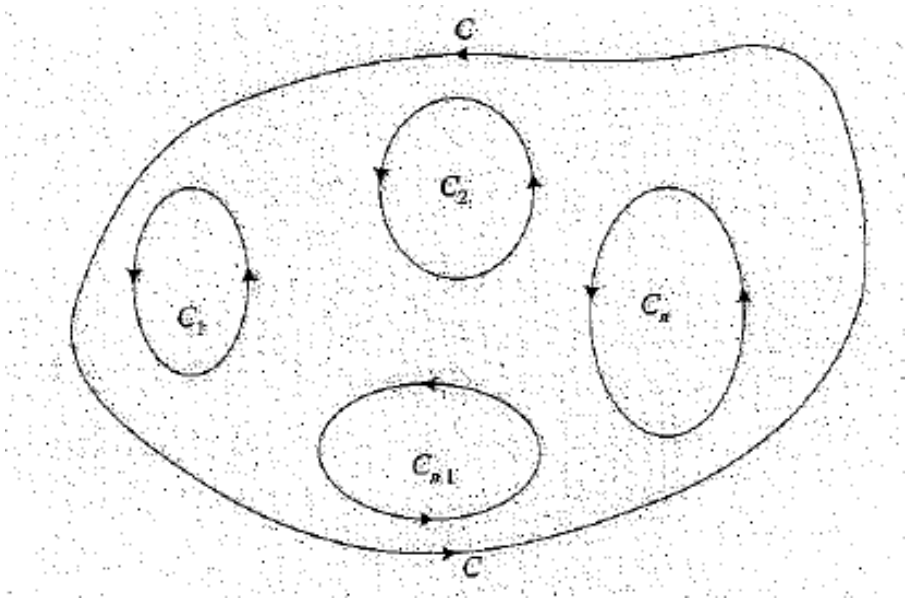
$$\oint_{C_1} f(z)\,dz - \oint_{C_2} f(z)\,dz = 0$$



# CAUCHY THEOREM for Multiply connected Region

The theorem can be extended to a multiply connected region bounded by an outer simple $C$ and inner simple closed curves $C_1, C_2, C_3, \ldots C_n$ as in the Figure 3.7. It can be proved that

$$\oint_C f(z)dz = \sum_{k=1}^{n} \oint_{C_k} f(z)dz,$$

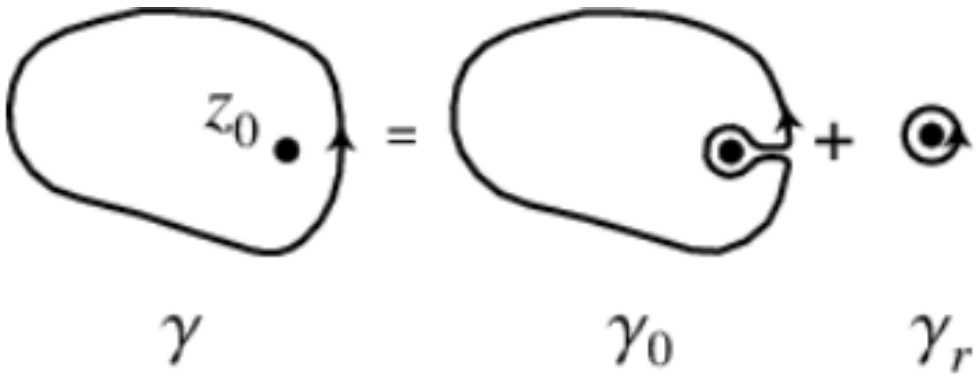


# Complex Integrals…

## Cauchy's Integral Formula

If $f(z)$ is analytic inside and on the boundary $C$ of a simply-connected region then for any point $z_0$ inside $C$,

$$\oint_C \frac{f(z)}{z - z_0}\, dz = 2\pi \mathrm{i}\ f(z_0).$$



## The derivative of an analytic function

If f(z) is analytic in a simply-connected region then at any interior point of the region, $z_0$ say, the derivatives of f(z) of any order exist and are themselves analytic. The derivatives at the point $z_0$ are given by Cauchy's integral formula for derivatives:

$$f^{(n)}(z_0) = \frac{n!}{2\pi \mathrm{i}} \oint_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz$$

**For every $n \in \mathbb{N}$**

where $C$ is any simple closed curve, in the region, which encloses $z_0$.
Note the case $n = 1$:

$$f'(z_0) = \frac{1}{2\pi \mathrm{i}} \oint_C \frac{f(z)}{(z - z_0)^2}\, dz.$$

# Examples

1. Find the contour Integral

$$\int_{|z|=3} \frac{e^z + \sin z}{z^2 - 16}\, dz,$$

2. If C is a unit circle, find

$$\int_C z^2\, dz$$

3. Evaluate $\int_i^{1+2i} \cos z\, dz$

4. Determine $\int_C \frac{6}{z(z-3)} dz$ ; where $C$ is the curve $|z-3| = 5$ .

5. $\oint_C \frac{e^z}{z^2-2z-3} dz$ ; $C$: $|z| \leq 2$

6. $\oint_C \frac{\cos z}{z^2-6z-5} dz$ ; $C$: $|z| \leq 4$

7. a) $\oint_C \frac{z}{(z+i)(z-i)} dz$ ; $C$: $|z-i| \leq \frac{1}{2}$

   b) $\oint_C \frac{z}{(z+i)(z-i)} dz$ ; $C$: $|z+i| \leq \frac{1}{2}$

   c) $\oint_C \frac{z}{(z+i)(z-i)} dz$ ; $C$: $|z-i| \leq 2$

8. a) $\oint_C \frac{z}{(z+3i)(z-3i)} dz$ ; $C1$: $|z-3i| \leq 1$

   b) $\oint_C \frac{z}{(z+3i)(z-3i)} dz$ ; $C1$: $|z+3i| \leq 1$

   c) $\oint_C \frac{z}{(z+3i)(z-3i)} dz$ ; $C1$: $|z| \leq 6$

# Examples

9. $\oint_C \frac{1}{z(z-1)}\,dz \;;\; C1\colon |z| \leq 2$

10. $\oint_C \frac{e^z}{z^3-z^2-5z-3}\,dz \;;\; C1\colon |z| \leq 2$

# Cauchy Residue Theorem

Suppose $f(z)$is a function which is analytic inside and on a closed contour $C$, except for a pole of order $m$at $z = z_0$ ,which lies inside $C$.

To evaluate

$$\oint_C f\,(z)\,dz,$$

we can expand $f(z)$in a Laurent series in powers of $(z - z_0)$.

If we let Γ be a circle with center $z_0$lying inside $C$, then,

$$\oint_C f\,(z)\,dz = \oint_\Gamma f\,(z)\,dz.$$

# Cauchy Residue Theorem

- we know that the integral of each of the positive and negative powers of $(z - z_0)$ is zero,

  with the exception of the term $\frac{b_1}{z-z_0}$, and this has the value $2\pi i\, b_1$.

- Since this is the only coefficient remaining after the integration, it is called the **residue** of $f(z)$at $z = z_0$ . It is given by

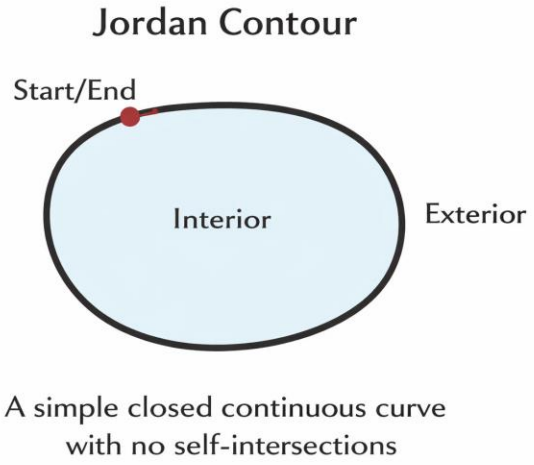
$$b_1 = \frac{1}{2\pi i} \oint_C f\,(z)\, dz.$$

- Calculating the residue for any given function $f(z)$ is an important task, and we examine some results concerning its determination for functions with simple poles, double poles, and poles of order $m$.

## Theorem

Suppose that the function f is analytic in a simply connected domain D, except for isolated singularities at z1, . . . , zk. Suppose further that C is a **Jordan contour** in D followed in the positive (anticlockwise) direction and not passing through z1, . . . , zk. Then

$$\frac{1}{2\pi \mathrm{i}} \int_C f(z)\,\mathrm{d}z = \sum_{\substack{j=1 \\ z_j \text{ inside } C}}^{k} \operatorname{res}(f, z_j).$$



# Finding Residue

If $f(z)$ has a simple pole at $z = z_0$ then $$f(z) = \frac{b_1}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \ldots$$

so that $$(z - z_0)f(z) = b_1 + a_0(z - z_0) + a_1(z - z_0)^2 + a_2(z - z_0)^3 + \ldots$$

Taking limits as $z \to z_0$, $$\lim_{z \to z_0} \{(z - z_0)f(z)\} = b_1.$$

In general the residue at a pole of order $m$ at $z = z_0$ is

$$b_1 = \frac{1}{(m-1)!} \lim_{z \to z_0} \left\{ \frac{d^{m-1}}{dz^{m-1}} \left[ (z - z_0)^m f(z) \right] \right\}.$$

# Examples

$$f(z) = \frac{1}{z^2+1} \equiv \frac{1}{(z+\mathrm{i})(z-\mathrm{i})} \equiv \frac{-\frac{1}{2i}}{z+\mathrm{i}} + \frac{\frac{1}{2\mathrm{i}}}{z-\mathrm{i}}.$$

There are simple poles at $z = -\mathrm{i}$ and $z = \mathrm{i}$. The residue at $z = \mathrm{i}$ is

$$\lim_{z\to\mathrm{i}} \left\{ (z-\mathrm{i}) \frac{1}{(z+\mathrm{i})(z-\mathrm{i})} \right\} = \lim_{z\to\mathrm{i}} \left( \frac{1}{z+\mathrm{i}} \right) = \frac{1}{2\mathrm{i}}.$$

Similarly, the residue at $z = -\mathrm{i}$ is

$$\lim_{z\to\mathrm{i}} \left\{ (z+\mathrm{i}) \frac{1}{(z+\mathrm{i})(z-\mathrm{i})} \right\} = \lim_{z\to-\mathrm{i}} \left( \frac{1}{z-\mathrm{i}} \right) = \frac{-1}{2\mathrm{i}}.$$

# Examples

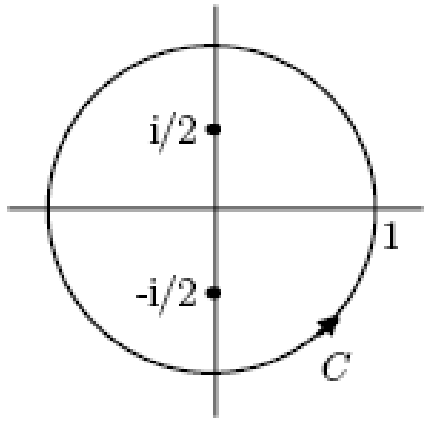
$f(z) = \dfrac{z^2+1}{(z+1)^3}$, $f(z)$ has a pole of order 3 at $z = -1$ $(m = 3)$.

We need first

$$\frac{d^2}{dz^2}\left[(z+1)^3\frac{(z^2+1)}{(z+1)^3}\right] = \frac{d^2}{dz^2}[z^2+1] = \frac{d}{dz}[2z] = 2.$$

Then $b_1 = \dfrac{1}{2!} \times 2 = 1.$

# Examples




$$f(z) = \frac{e^{2iz}}{1+4z^2}$$

has simple poles at $z = \pm i/2$, with residues

$$\operatorname{res}\left(f, \frac{i}{2}\right) = \lim_{z\to i/2}\left(z - \frac{i}{2}\right) f(z) = \lim_{z\to i/2} \frac{e^{2iz}}{4(z+i/2)} = \frac{e^{-1}}{4i}$$

and

$$\operatorname{res}\left(f, -\frac{i}{2}\right) = \lim_{z\to -i/2}\left(z + \frac{i}{2}\right) f(z) = \lim_{z\to -i/2} \frac{e^{2iz}}{4(z-i/2)} = -\frac{e}{4i}.$$

It follows from Cauchy's residue theorem that if $C = \{z : |z| = 1\}$ is the circle with centre 0 and radius 1, followed in the positive (anticlockwise) direction, then

$$\int_C \frac{e^{2iz}}{1+4z^2}\,dz = 2\pi i\left(\frac{e^{-1}}{4i} - \frac{e}{4i}\right) = \frac{\pi}{2}\left(\frac{1}{e} - e\right).$$

# Examples

$$f(z) = \frac{e^z}{z^4}$$

has a pole of order 4 at $z = 0$, with residue

$$\operatorname{res}(f, 0) = \frac{1}{3!} \lim_{z \to 0} \frac{d^3}{dz^3}(z^4 f(z)) = \frac{1}{3!} \lim_{z \to 0} \frac{d^3}{dz^3} e^z = \frac{1}{6}.$$

It follows from Cauchy's residue theorem that if $C$ is any Jordan contour with 0 inside and followed in the positive (anticlockwise) direction, then

$$\int_C \frac{e^z}{z^4}\, dz = 2\pi i \left(\frac{1}{6}\right) = \frac{\pi i}{3}.$$